MIPT & YSDA · AUTUMN 2026

Deep Generative Models

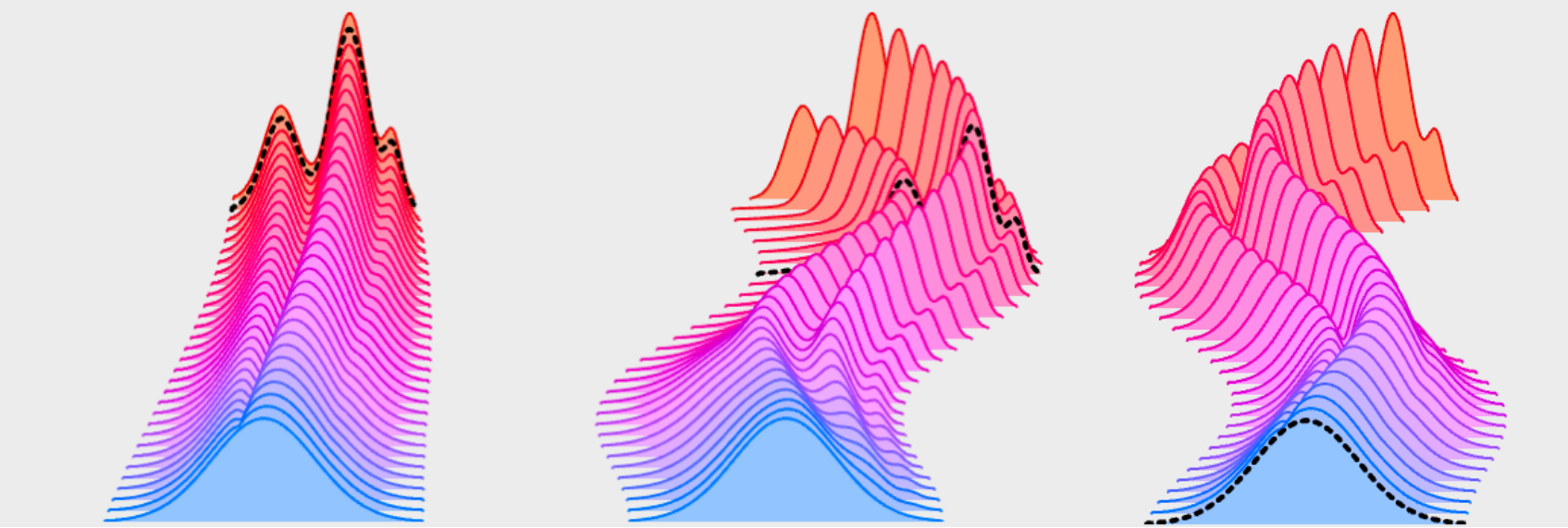
Lecture 12

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Recap of Previous Lecture

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$



Recap of Previous Lecture

Problem: The true vector field $\mathbf{v}(\mathbf{x}, t)$ is **unknown**.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})\right).$$

Recap of Previous Lecture

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

If $p_t(\mathbf{x})>0$ for all $\mathbf{x}\in\mathbb{R}^m$ and $0\leq t<1$, the FM and CFM objectives differ by a constant independent of $\boldsymbol{\theta}$, so they have the same minimizers.

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

Recap of Previous Lecture

Training

1. Sample $t \sim U[0, 1]$, $\mathbf{z} \sim p(\mathbf{z})$, $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
2. Compute loss $\mathcal{L} = \|\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \boldsymbol{\psi}_1(\mathbf{x}_0) = \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1).$$

Recap of Previous Lectures

Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t)\frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})}d\mathbf{z}$$

Flow Matching (FM)

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 \to \min_{\boldsymbol{\theta}}$$

Recap of Previous Lectures

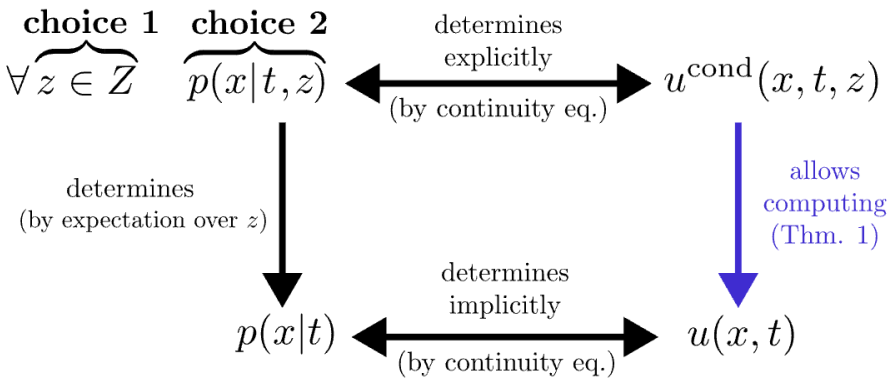
Conditional Flow Matching (CFM)

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Theorem

If $p_t(\mathbf{x})>0$ for all $\mathbf{x}\in\mathbb{R}^m$ and $0\leq t<1$, the FM and CFM objectives differ by a constant independent of $\boldsymbol{\theta}$, so they have the same minimizers.

Recap of Previous Lecture



Open Questions

- **Q1** How should we choose the convenient conditioning latent variable $\mathbf{z}$?
- **Q2** How can we parametrize $p_t(\mathbf{x})$ (or $p_t(\mathbf{x}|\mathbf{z})$) to enforce the following constraints?

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

image credit: A Visual Dive into Conditional Flow Matching

Recap of Previous Lecture

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right);$$
$$\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

image credit: A Visual Dive into Conditional Flow Matching

Recap of Previous Lecture

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right);$$
$$\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

$$\begin{aligned}\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) &= \boldsymbol{\psi}_t'\left(\boldsymbol{\psi}_t(\mathbf{x}_t, \mathbf{z})^{-1}, \mathbf{z}\right) \\ &= \boldsymbol{\mu}_t'(\mathbf{z}) + \frac{\boldsymbol{\sigma}_t'(\mathbf{z})}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))\end{aligned}$$

Recap of Previous Lecture

Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z} = \mathbf{x}_1$. Then $p(\mathbf{z}) = p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{x}_1)p_1(\mathbf{x}_1)d\mathbf{x}_1$$

We need to ensure the boundary constraints:

$$\begin{cases} \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}) \\ \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}) \end{cases} \Rightarrow \quad \begin{cases} p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \\ p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1) \end{cases}$$

Recap of Previous Lecture

$$p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \quad p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_1)\right);$$
$$\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases} \Rightarrow \begin{cases} p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2 \cdot \mathbf{I}\right) \\ \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \end{cases}$$

image credit: A Visual Dive into Conditional Flow Matching

Recap of Previous Lecture

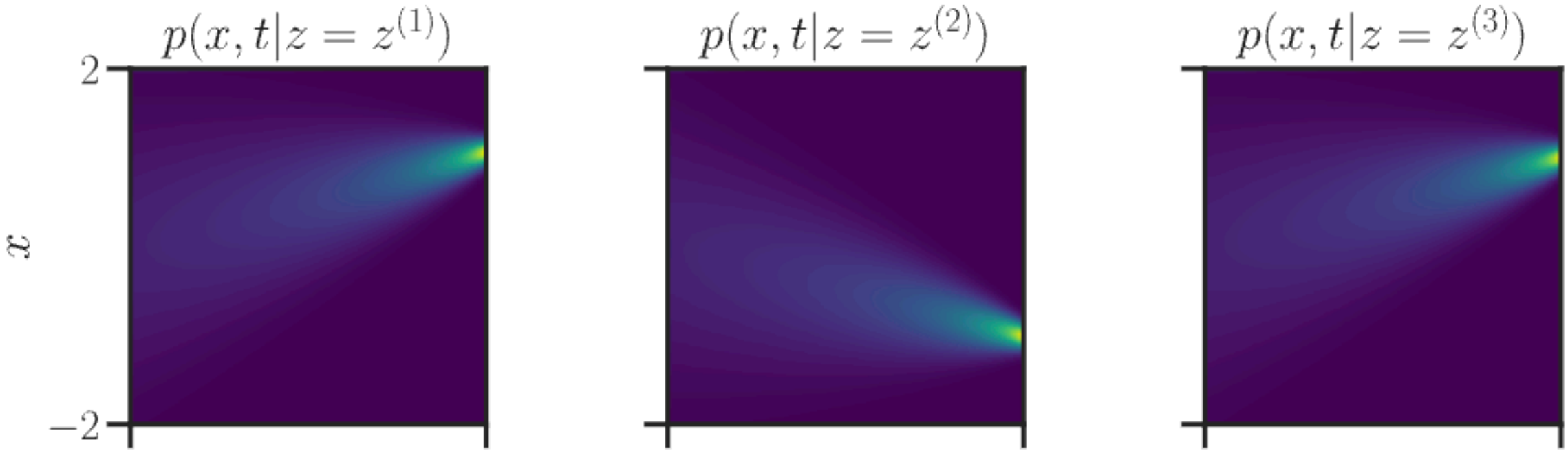


image credit: A Visual Dive into Conditional Flow Matching

Recap of Previous Lecture

Rectified Flow Objective

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2\mathbf{I}\right); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

$$\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1-t} = \mathbf{x}_1 - \mathbf{x}_0, \qquad t < 1$$

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left\|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2$$

$$= \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})} \left\|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\boldsymbol{\theta}}\left(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, t\right)\right\|^2$$

At $t = 1$, the quotient is undefined. Use the simplified training target $\mathbf{x}_1 - \mathbf{x}_0$, which contains no division.

image credit: https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html
Liu X., Gong C., Liu Q. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow, 2022

Recap of Previous Lecture

- $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t)$ defines straight lines between $p_{\text{data}}(\mathbf{x})$ and $\mathcal{N}(0, \mathbf{I})$.
- **Straight training paths do not imply straight sampling trajectories.**



image credit: https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html
Liu X., Gong C., Liu Q. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow, 2022

Recap of Previous Lecture

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})}\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Training

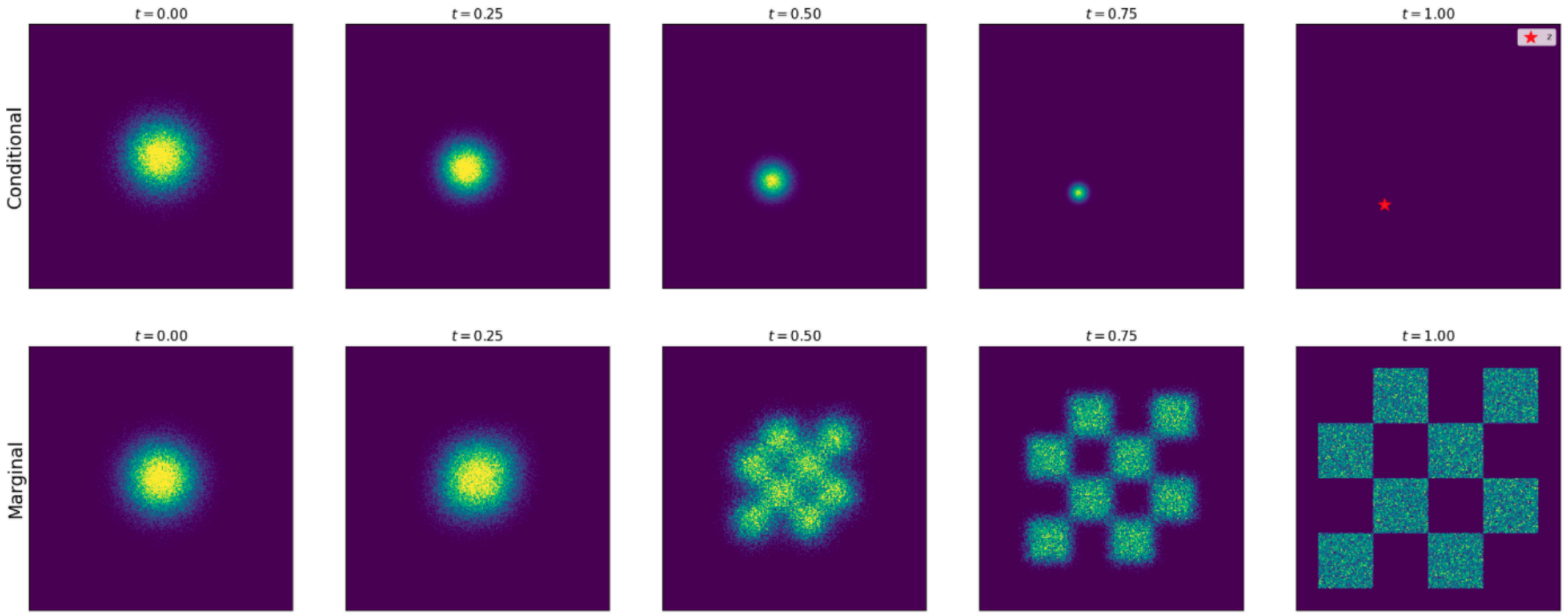
1. Sample $\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})$, $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$, $t\sim U[0,1]$.
2. Compute noisy image $\mathbf{x}_t=t\mathbf{x}_1+(1-t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L}=\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\boldsymbol{\psi}_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

Recap of Previous Lecture



- One-sided conditioning gives us the way to construct a generative model.
- Now we extend it to image-to-image formulation (mapping between two distinct data distributions $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$).

Recap of Previous Lecture

Rectified Flow: Training-Time Distribution

Keep $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$ and the target $\mathbf{x}_1 - \mathbf{x}_0$ fixed. Let $\ell_{\boldsymbol{\theta}}(t)$ be the expected velocity MSE at time t.

$$\mathbb{E}_{t\sim r}\ell_{\boldsymbol{\theta}}(t) = \mathbb{E}_{t\sim U[0,1]}\left[r(t)\ell_{\boldsymbol{\theta}}(t)\right].$$

Changing the time distribution without correction changes the objective. To preserve uniform-time training, use $\ell_{\boldsymbol{\theta}}(t)/r(t)$ with $t \sim r$.

Logit-Normal Example

$$u \sim \mathcal{N}(\mu, s^2), \qquad t = \frac{1}{1+e^{-u}}, \qquad \log \mathrm{SNR}(t) = 2\log\frac{t}{1-t} = 2u.$$

Time $t=0$ is noise and $t=1$ is data. This choice does not prescribe the ODE grid.

Outline

01 Two-Sided Conditioning

02 Link between Flow Matching and Score-Based Models

03 Discrete Diffusion Models

04 Forward Discrete Process

05 Reverse Diffusion Process

06 From Token To Sequence

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Pair Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z}=(\mathbf{x}_0,\mathbf{x}_1)$. Then $p(\mathbf{z})=p(\mathbf{x}_0,\mathbf{x}_1)=p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x})=\int p_t(\mathbf{x}|\mathbf{x}_0,\mathbf{x}_1)p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)d\mathbf{x}_0d\mathbf{x}_1$$

Pair Conditioning

We must enforce boundary constraints:

$$\begin{cases} p_0(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}); \\ p_1(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) \end{cases} \Rightarrow \quad \begin{cases} p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0) \\ p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1) \end{cases}$$

Two-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)\right);$$

$$\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

image credit: A Visual Dive into Conditional Flow Matching

Two-Sided Conditioning

Let’s consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \\ \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) = \epsilon \end{cases} \Rightarrow \begin{cases} p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0) \\ p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1) \end{cases}$$

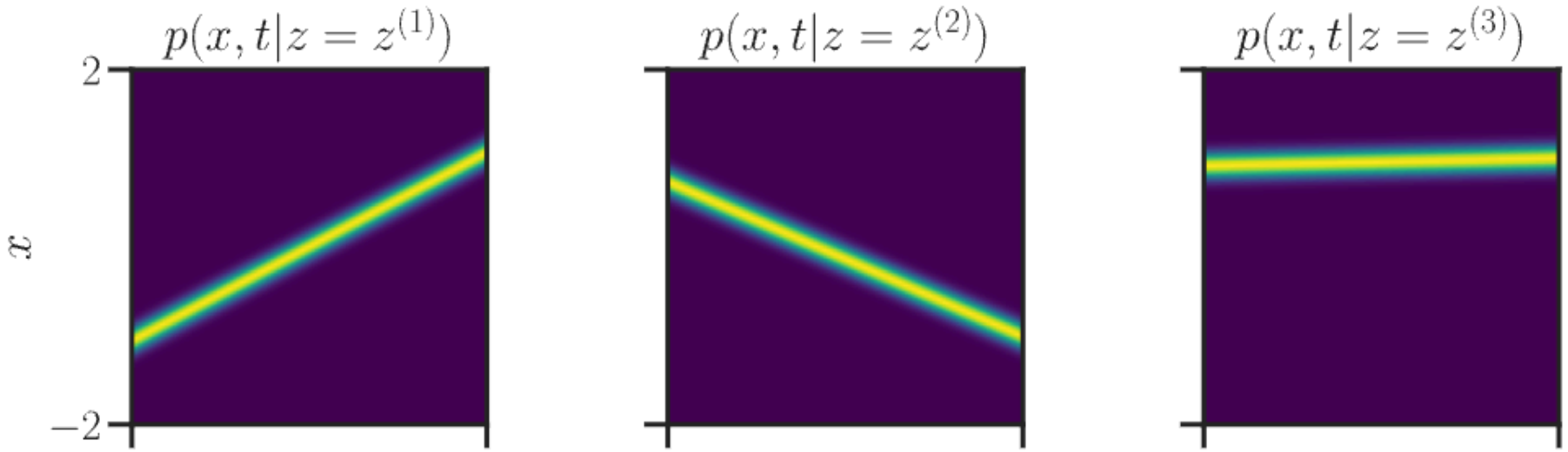
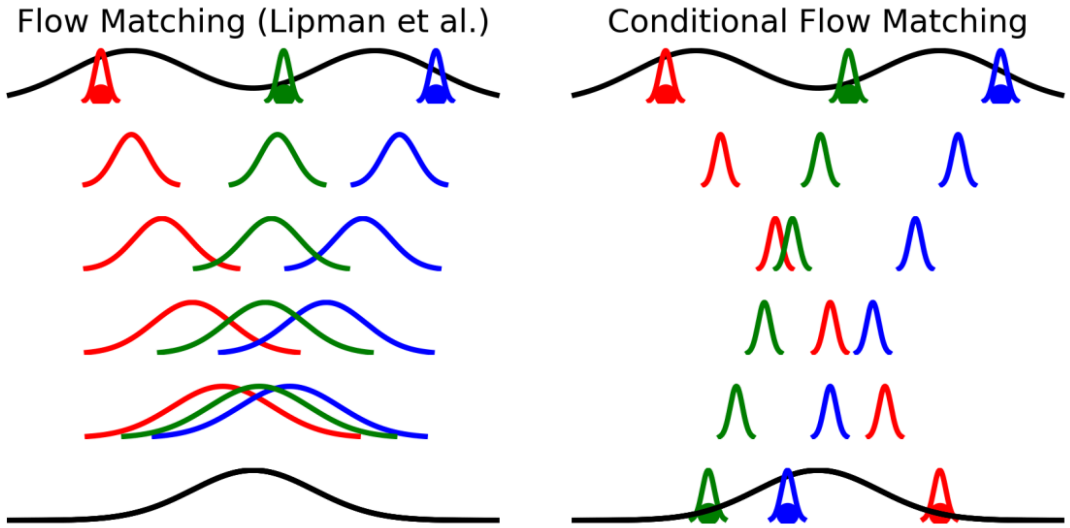


image credit: A Visual Dive into Conditional Flow Matching

Flow Matching: One-Sided vs. Two-Sided Conditioning

$$\mathbf{z} = \mathbf{x}_1$$
$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2\mathbf{I}\right)$$
$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

$$\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$$
$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2\mathbf{I}\right)$$
$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$



Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2\mathbf{I}\right);$$
$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\begin{aligned}\frac{d\mathbf{x}_t}{dt} &= \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) \\ &= \boldsymbol{\mu}'_t(\mathbf{x}_0, \mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_0, \mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1))\end{aligned}$$

$$\mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) = \mathbf{x}_1 - \mathbf{x}_0$$

Two-Sided Conditioning: Vector Field and Objective

Conditional Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$
$$=\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{(\mathbf{x}_0,\mathbf{x}_1)\sim p(\mathbf{x}_0,\mathbf{x}_1)}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{x}_0,\mathbf{x}_1)}\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

- This yields the same procedure as for one-sided conditioning!
- Now, we do not require that $p_0(\mathbf{x})$ is necessarily $\mathcal{N}(0,\mathbf{I})$.

CFM with Two-Sided Conditioning: Training and Sampling

- This conditioning allows us to transport any distribution $p_0(\mathbf{x})$ to any distribution $p_1(\mathbf{x})$.
- It's possible to apply this approach to paired tasks, e.g., style transfer.

Training

1. Sample $(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)$, $t \sim U[0, 1]$.
2. Compute noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\|^2$.

CFM with Two-Sided Conditioning: Training and Sampling

Sampling

1. Sample $\mathbf{x}_0 \sim p_0(\mathbf{x})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \boldsymbol{\psi}_1(\mathbf{x}_0) = \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1).$$

Stable Diffusion 3: Scalable Flow Matching



Training connection: SD3 combines rectified flow with nonuniform training times, including logit-normal sampling.

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01 Two-Sided Conditioning

02 Link between Flow Matching and Score-Based Models

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Score-Based Generative Models through SDEs

Training Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t)-\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$

Variance Exploding SDE (NCSN)

$$q(\mathbf{x}(t)|\mathbf{x}(0))=\mathcal{N}\left(\mathbf{x}(0),\left[\sigma^2(t)-\sigma^2(0)\right]\cdot\mathbf{I}\right),\quad \sigma(0)=0$$

Variance Preserving SDE (DDPM)

$$q(\mathbf{x}(t)|\mathbf{x}(0))=\mathcal{N}\left(\mathbf{x}(0)\alpha(t),\left(1-\alpha(t)^2\right)\cdot\mathbf{I}\right);$$
$$\alpha(t)=e^{-\frac{1}{2}\int_0^t\beta(s)ds}$$

Score-Based Generative Models through SDEs

Flow matching uses reverse time direction:

$$p_t(\mathbf{x}|\mathbf{x}_1) = q_{1-t}(\mathbf{x}|\mathbf{x}_0 = \mathbf{x}_1)$$

Tweedie's Formula

Recall Lecture 6, now with a scaled signal: for $\alpha_t > 0$, $\sigma_t > 0$,

$$p_t(\mathbf{x}_t|\mathbf{x}_1) = \mathcal{N}(\alpha_t \mathbf{x}_1, \sigma_t^2 \mathbf{I}) \quad \Rightarrow \quad \mathbb{E}[\mathbf{x}_1 \mid \mathbf{x}_t] = \frac{\mathbf{x}_t + \sigma_t^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)}{\alpha_t}.$$

When the conditional velocity is affine in $\mathbf{x}_1$, averaging over $\mathbf{x}_1 \mid \mathbf{x}_t$ expresses the marginal velocity through the score.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022
Efron B. Tweedie's Formula and Selection Bias, 2011

Score-Based Generative Models through SDEs

$$p_t(\mathbf{x}|\mathbf{x}_1) = q_{1-t}(\mathbf{x}|\mathbf{x}_0 = \mathbf{x}_1)$$

$$\textbf{VE (NCSN): } p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\mathbf{x}_1, \sigma^2_{1-t} \cdot \mathbf{I}\right)$$

$$\textbf{VP (DDPM): } p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\alpha_{1-t}\mathbf{x}_1, \left(1 - \alpha^2_{1-t}\right) \cdot \mathbf{I}\right)$$

Score-Based Generative Models through SDEs

Flow Matching Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2\mathbf{I}\right); \quad \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1-t}$$

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \boldsymbol{\mu}_t'(\mathbf{x}_1) + \frac{\boldsymbol{\sigma}_t'(\mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_1))$$

Let’s derive the conditional vector fields for VE (NCSN) and VP (DDPM).

Flow Matching vs Score-Based SDE Models

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \boldsymbol{\mu}'_t(\mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_1))$$

Variance Exploding SDE Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\mathbf{x}_1, \sigma_{1-t}^2 \mathbf{I}\right) \quad \Rightarrow \quad \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = -\frac{\sigma'_{1-t}}{\sigma_{1-t}}(\mathbf{x}_t - \mathbf{x}_1)$$

Variance Preserving SDE Probability Path

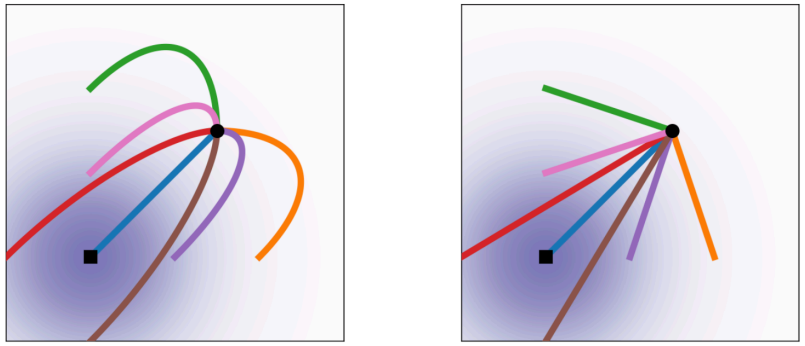
$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\alpha_{1-t}\mathbf{x}_1, (1 - \alpha_{1-t}^2)\mathbf{I}\right)$$

$$\Rightarrow \quad \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\alpha'_{1-t}}{1 - \alpha_{1-t}^2} \cdot (\alpha_{1-t}\mathbf{x}_t - \mathbf{x}_1)$$

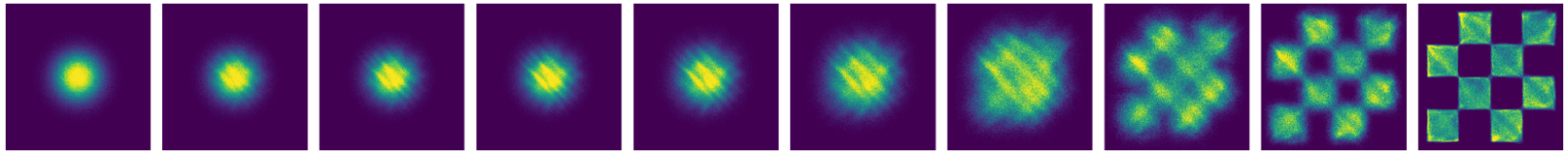
Thus, VE/VP SDE models correspond to particular choices of the Gaussian probability path within the flow matching framework.

Flow Matching vs Score-Based SDE Models

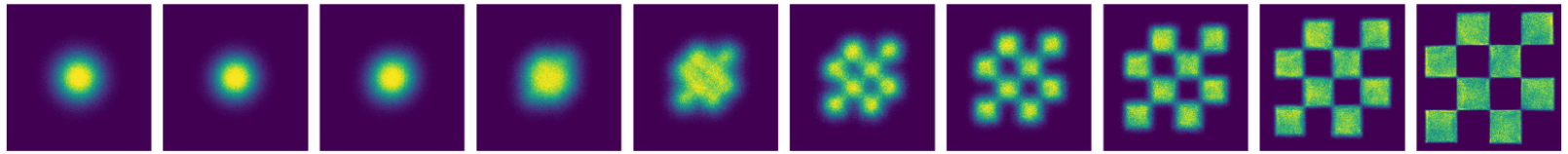
Trajectories



Diffusion OT



Score matching w/ Diffusion



Flow Matching w/ OT

Outline

01 Two-Sided Conditioning

02 Link between Flow Matching and Score-Based Models

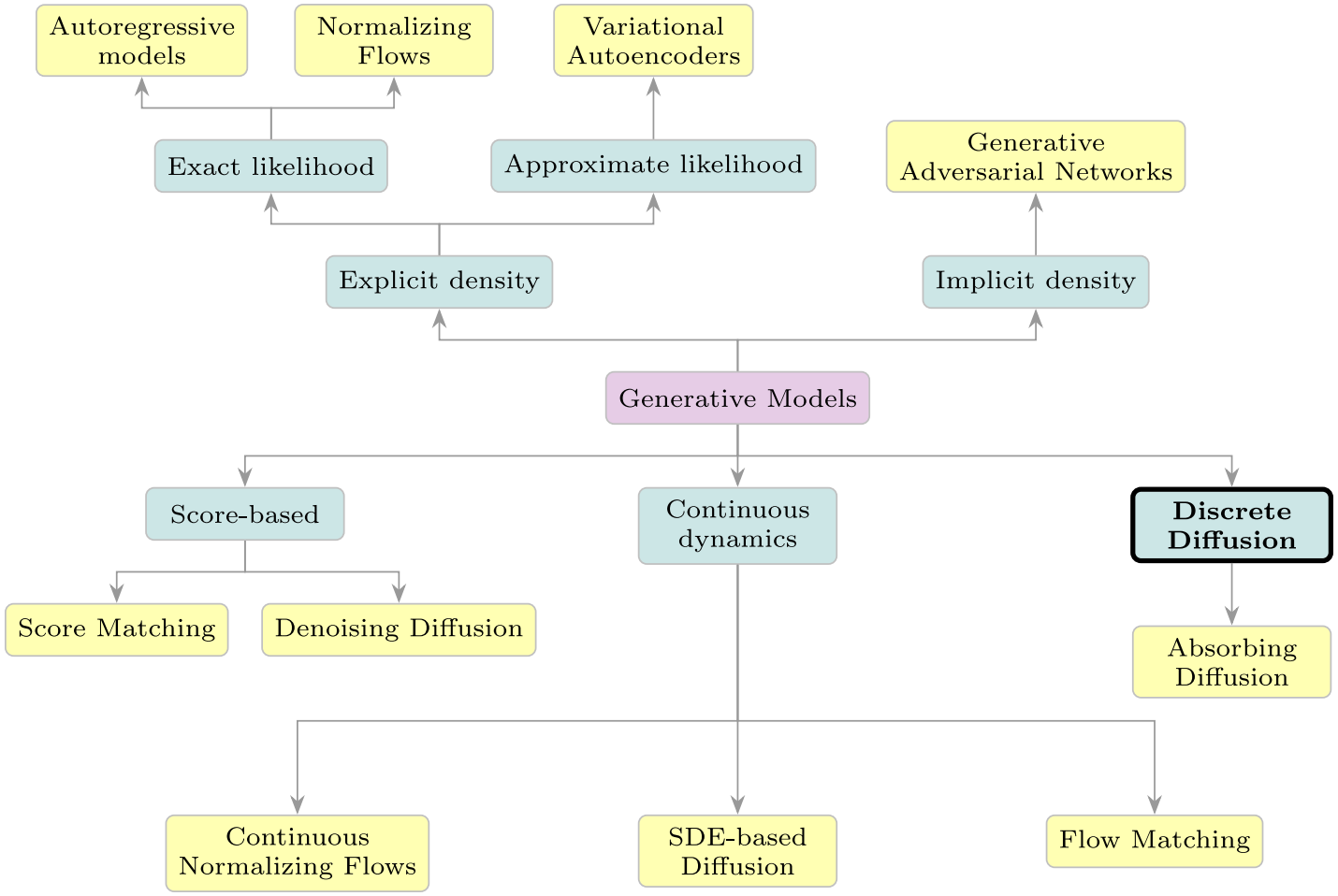
03 Discrete Diffusion Models

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Generative Models Taxonomy



Discrete or Continuous Diffusion Models?

Reminder: Diffusion models define a forward corruption process and a reverse denoising process. Previously, we studied diffusion models with continuous states $\mathbf{x}(t) \in \mathbb{R}^m$.

Continuous state space

- **Discrete time** $t \in \{0, 1, \dots, T\}$ $\Rightarrow$ **DDPM / NCSN**.
- **Continuous time** $t \in [0, 1]$ $\Rightarrow$ **Score-based SDE models**.

Now we turn to diffusion over discrete-value states $\mathbf{x}(t) \in \{1, \dots, K\}^m$.

Discrete state space

- **Discrete time** $t \in \{0, 1, \dots, T\}$.
- **Continuous time** $t \in [0, 1]$.

Let's discuss why we need discrete diffusion models.

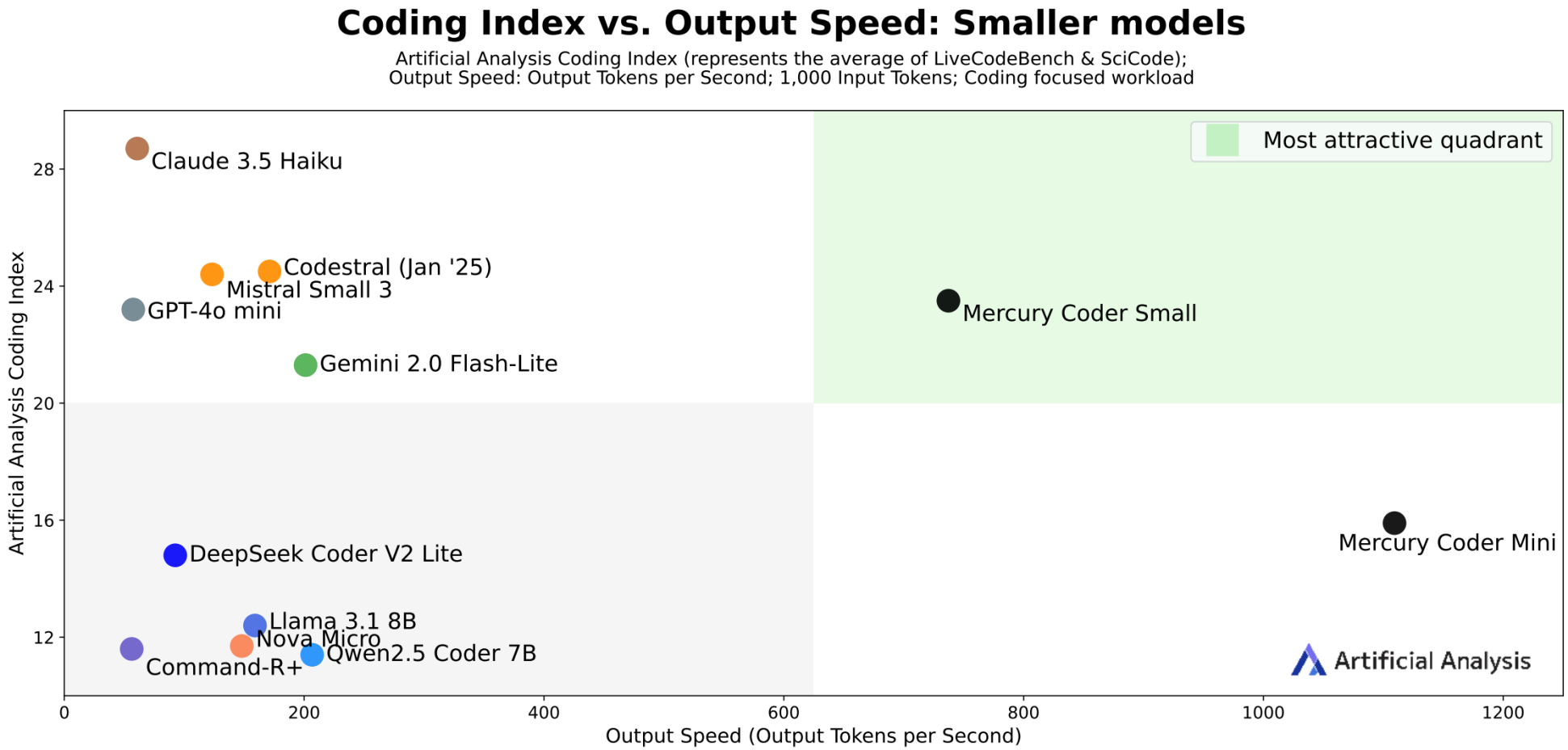
Why Discrete Diffusion Models?

Potential advantages over left-to-right autoregressive generation

- **Parallel generation:** generate multiple tokens per step, potentially reducing sequential decoding steps.
- **Bidirectional context:** use available context on both sides when predicting a token.
- **Flexible generation order:** choose which positions to generate next, including confidence-based ordering.
- **Infilling and controllability:** fill arbitrary gaps, preserve specified tokens, and steer sampling with guidance.
- **Iterative refinement:** revisit generated tokens when the sampler supports remasking or corrective updates.

Nie S. et al. Large language diffusion models, 2025. Wu C. et al. Fast-dLLM, 2025.
Schiff Y. et al. Simple guidance mechanisms for discrete diffusion models, 2025. Wang G. et al. Remasking discrete diffusion models with inference-time scaling, 2025.

2025 – Big Bang of Discrete Diffusion Models



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Forward Discrete Process

Continuous Diffusion Markov Chain

In continuous diffusion, the forward Markov chain is defined by progressively corrupting data with Gaussian noise:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}).$$

Discrete Diffusion Markov Chain

For discrete data, we instead define a Markov chain over categorical states:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathrm{Cat}(\mathbf{Q}_t\mathbf{x}_{t-1}),$$

- Each $\mathbf{x}_t \in \{0,1\}^K$ is a **one-hot vector** encoding the categorical state (it is just one token).
- What is the transition matrix $\mathbf{Q}_t$?

Forward Process over Time

Transition Matrix

$\mathbf{Q}_t \in [0,1]^{K \times K}$ is a **transition matrix** where each column gives transition probabilities from one state to all others, and columns sum to 1:

$$[\mathbf{Q}_t]_{ij} = q(x_t = i | x_{t-1} = j), \qquad \sum_{i=1}^{K} [\mathbf{Q}_t]_{ij} = 1.$$

- The forward diffusion gradually destroys information through repeated random transitions.
- Applying the transition t times yields the marginal distribution:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathrm{Cat}(\mathbf{Q}_{1:t}\mathbf{x}_0), \qquad \mathbf{Q}_{1:t} = \mathbf{Q}_t \mathbf{Q}_{t-1} \cdots \mathbf{Q}_1.$$

- As $t \to T$, the process drives the data toward a stationary distribution.
- We design the transition matrices $\mathbf{Q}_t$ to achieve this behavior.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Transition Matrix

- The choice of $\mathbf{Q}_t$ determines how information is erased and what the stationary distribution becomes.
- $\mathbf{Q}_t$ and $\mathbf{Q}_{1:t}$ should be easy to compute for each t.

Common choices

- **Uniform diffusion**

  $$\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t \mathbf{U}, \qquad \mathbf{U}_{ij} = \frac{1}{K}.$$

  Each token is replaced by a uniformly random symbol with probability β_t. The stationary distribution is uniform noise.

- **Absorbing diffusion**

  $$\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t \, \mathbf{e}_m \mathbf{1}^\top.$$

  Tokens are gradually replaced by a special mask m; the stationary distribution is fully masked.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Transition Matrix

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathrm{Cat}(\mathbf{Q}_{1:t}\mathbf{x}_0), \qquad \mathbf{Q}_{1:t} = \mathbf{Q}_t\mathbf{Q}_{t-1}\cdots\mathbf{Q}_1.$$

Uniform Diffusion

$$\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t\mathbf{U}, \qquad \mathbf{U}_{ij} = \frac{1}{K}.$$

$$\mathbf{Q}_{1:t} = \bar{\alpha}_t\mathbf{I} + (1-\bar{\alpha}_t)\mathbf{U}, \quad \bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s).$$

- Each token retains its original value with prob. $\bar{\alpha}_t$.
- It becomes uniformly random with prob. $(1-\bar{\alpha}_t)$.
- As $t \to T$, the process converges to the stationary uniform distribution.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Transition Matrix

Absorbing Diffusion

$$\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t\,\mathbf{e}_m\mathbf{1}^\top,$$

$$\mathbf{Q}_{1:t} = \bar{\alpha}_t\,\mathbf{I} + (1-\bar{\alpha}_t)\,\mathbf{e}_m\mathbf{1}^\top, \qquad \bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s).$$

- Each token retains its original value with prob. $\bar{\alpha}_t$.
- It becomes $\mathbf{e}_m$ with prob. $(1-\bar{\alpha}_t)$.
- As $t \to T$, all tokens converge to the mask state: $q(\mathbf{x}_T) \approx \mathrm{Cat}(\mathbf{e}_m)$.
- This makes the process analogous to **masked language modeling**.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Uniform vs. Absorbing Transition Matrix

Aspect	Uniform Diffusion	Absorbing Diffusion
$\mathbf{Q}_t$	$(1-\beta_t)\mathbf{I} + \beta_t \mathbf{U}$	$(1-\beta_t)\mathbf{I} + \beta_t \mathbf{e}_m \mathbf{1}^\top$
$\mathbf{Q}_{1:t}$	$\bar{\alpha}_t \mathbf{I} + (1-\bar{\alpha}_t)\mathbf{U}$	$\bar{\alpha}_t \mathbf{I} + (1-\bar{\alpha}_t)\mathbf{e}_m \mathbf{1}^\top$
$\mathbf{Q}_{1:\infty}$	$\mathbf{U}$	$\mathrm{Cat}(\mathbf{e}_m)$
Interpretation	Random replacement	Gradual masking of tokens
Application	Image diffusion	Text diffusion $\approx$ Masked LM

Observation

Both schemes gradually destroy information, but differ in their stationary limit. Absorbing diffusion bridges diffusion and masked-language-model objectives.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

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Posterior of the Forward Process

ELBO

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)\|p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$

- Conditioned reverse distribution $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ played crucial role in the continuous-state diffusion model.
- It shows the probability of a previous state given the noisy state $\mathbf{x}_t$ and the original clean data $\mathbf{x}_0$.

Posterior of the Forward Process

Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)\, q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$
$$= \frac{\mathrm{Cat}(\mathbf{Q}_t) \cdot \mathrm{Cat}(\mathbf{Q}_{1:t-1})}{\mathrm{Cat}(\mathbf{Q}_{1:t})}.$$

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Posterior of the Forward Process

Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\frac{\mathbf{Q}_t^\top \mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right).$$

Recall the ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big),$$

- Both $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $q(\mathbf{x}_t|\mathbf{x}_0)$ are known analytically from the forward process.
- The reverse process $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is a learned categorical distribution:

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}\big(\boldsymbol{\pi}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\big),$$

where $\boldsymbol{\pi}_{\boldsymbol{\theta}}$ is a neural network.

Discrete-time ELBO for Discrete Diffusion

ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big).$$

Categorical KL

$$D_{\mathrm{KL}}\big(\mathrm{Cat}(\mathbf{q}) \,\|\, \mathrm{Cat}(\mathbf{p})\big) = \sum_{k=1}^{K} q_k \log \frac{q_k}{p_k} = \mathrm{H}(\mathbf{q}, \mathbf{p}) - \mathrm{H}(\mathbf{q}),$$

- $\mathrm{H}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)\big)$ is a constant w.r.t. $\boldsymbol{\theta}$.
- $\mathrm{H}(\mathbf{q}, \mathbf{p}) = -\sum_k q_k \log p_k$ is a **cross-entropy loss**.

Therefore, minimizing $\mathcal{L}_t$ w.r.t. $\boldsymbol{\theta}$ is equivalent to minimizing

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \mathrm{H}\Big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0),\; p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\Big).$$

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

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From Token to Sequence

One-hot sequence representation

$$\mathbf{x}_t \in \{0,1\}^K \quad \Leftrightarrow \quad \mathbf{X}_t \in \{0,1\}^{K \times m}$$

Here $\mathbf{X}_t$ is a one-hot representation of a sequence of tokens.

Independent Token-wise Forward Process

$$q(\mathbf{X}_t|\mathbf{X}_{t-1}) = \prod_{i=1}^{m} q(\mathbf{x}_t^i|\mathbf{x}_{t-1}^i) = \mathrm{Cat}(\mathbf{Q}_t\mathbf{X}_{t-1})$$

- Each position i evolves according to its own Markov chain.
- Often the same transition matrix $\mathbf{Q}_t$ is shared across i.

Continuous Diffusion Analogy

- In Gaussian DDPMs with diagonal covariance, noise is independent per pixel.
- Structure is not in the noise; it is learned by the reverse model.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

From Token to Sequence

$$q(\mathbf{X}_t|\mathbf{X}_{t-1}) = \prod_{i=1}^{m} q(\mathbf{x}_t^i|\mathbf{x}_{t-1}^i) = \mathrm{Cat}(\mathbf{Q}_t\mathbf{X}_{t-1})$$

$$q(\mathbf{X}_t|\mathbf{X}_0) = \mathrm{Cat}(\mathbf{Q}_{1:t}\mathbf{X}_0)$$

Conditioned Reverse Distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathrm{Cat}\left(\frac{\mathbf{Q}_t^\top \mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right).$$

$$q(\mathbf{X}_{t-1}|\mathbf{X}_t, \mathbf{X}_0) = \prod_{i=1}^{m} q(\mathbf{x}_{t-1}^i|\mathbf{x}_t^i, \mathbf{x}_0^i).$$

- All distributions defined by the forward process are factorized.
- Dependence appears in the learned reverse model.

Reverse Model for Sequence

$$p_{\boldsymbol{\theta}}(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

- The output factorizes (parallel prediction across positions).
- Each factor conditions on the entire noisy sequence $\mathbf{X}_t$.
- This is exactly the **masked language modeling** pattern.

Objective: $\mathcal{L}_t$ term

$$D_{\mathrm{KL}}\left(q(\mathbf{X}_{t-1}|\mathbf{X}_t, \mathbf{X}_0) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{X}_{t-1}|\mathbf{X}_t)\right)$$
$$= \sum_{i=1}^{m} D_{\mathrm{KL}}\left(q(\mathbf{x}_{t-1}^i|\mathbf{x}_t^i, \mathbf{x}_0^i) \| p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}^i|\mathbf{X}_t)\right).$$

Reverse Model for Sequence

Final objective: masked LM

$$\mathcal{L} = \sum_{t=1}^{T} \sum_{i=1}^{m} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \Big[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_t) \Big].$$

Summary

- Two-sided conditioning uses pair conditioning $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$ and yields the same procedure, but is more general (suitable for paired tasks).
- Diffusion and score-based models are special cases of the flow matching approach, but use curved trajectories.
- Discrete diffusion offers parallel generation and flexible conditioning; token revision requires a suitable sampler.
- Forward discrete diffusion process defines a Markov chain with discrete states; uniform and absorbing transitions make it tractable.
- Reverse discrete diffusion process uses the variational approach to invert forward process; the discrete-state ELBO is a cross-entropy loss.
- For sequences, the forward process factorizes over tokens, while the learned reverse model predicts all positions in parallel conditioning on the whole noisy sequence (masked LM pattern).